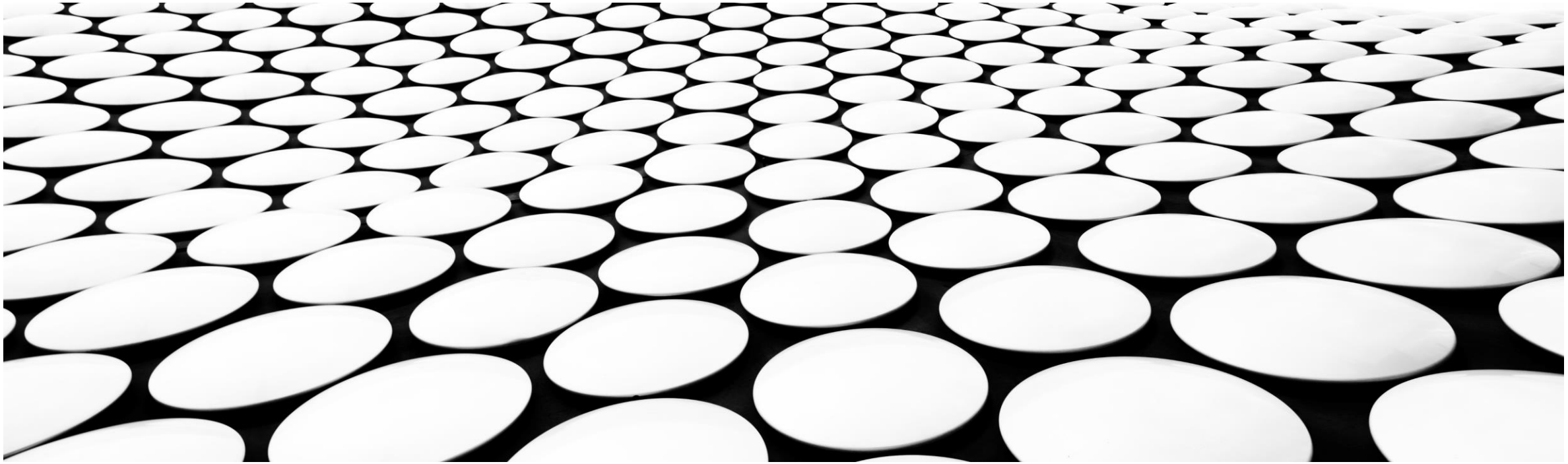


Engineering Security Solutions

Ransomware – Rollback Remediation

Donnacha Forde



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$whoami

- Developer
- Architect
- Manager
- Technical Director

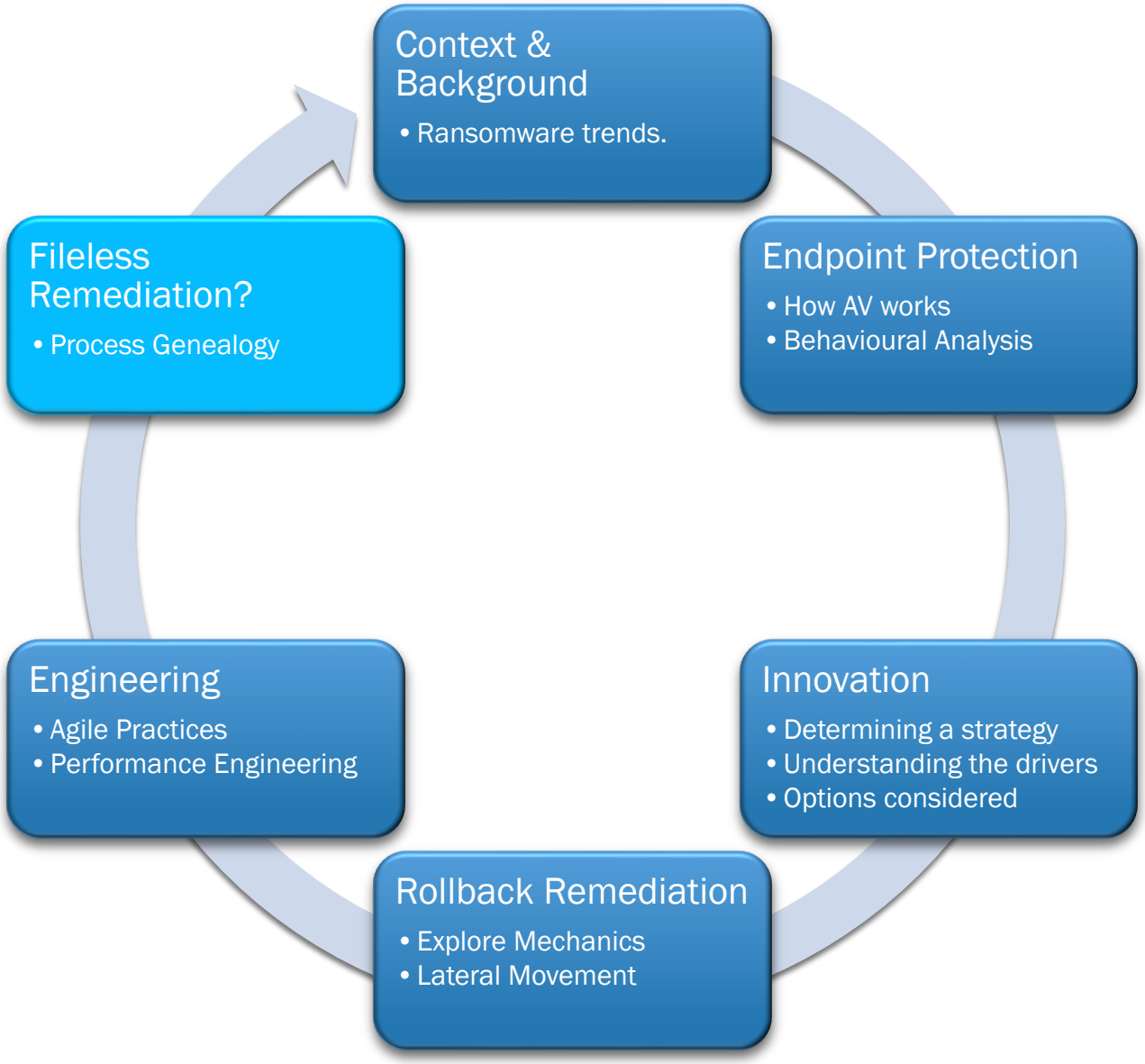


Mostly Finance (~10yrs)

Mostly Systems-Management (~15yrs)

Security (>8 yrs)

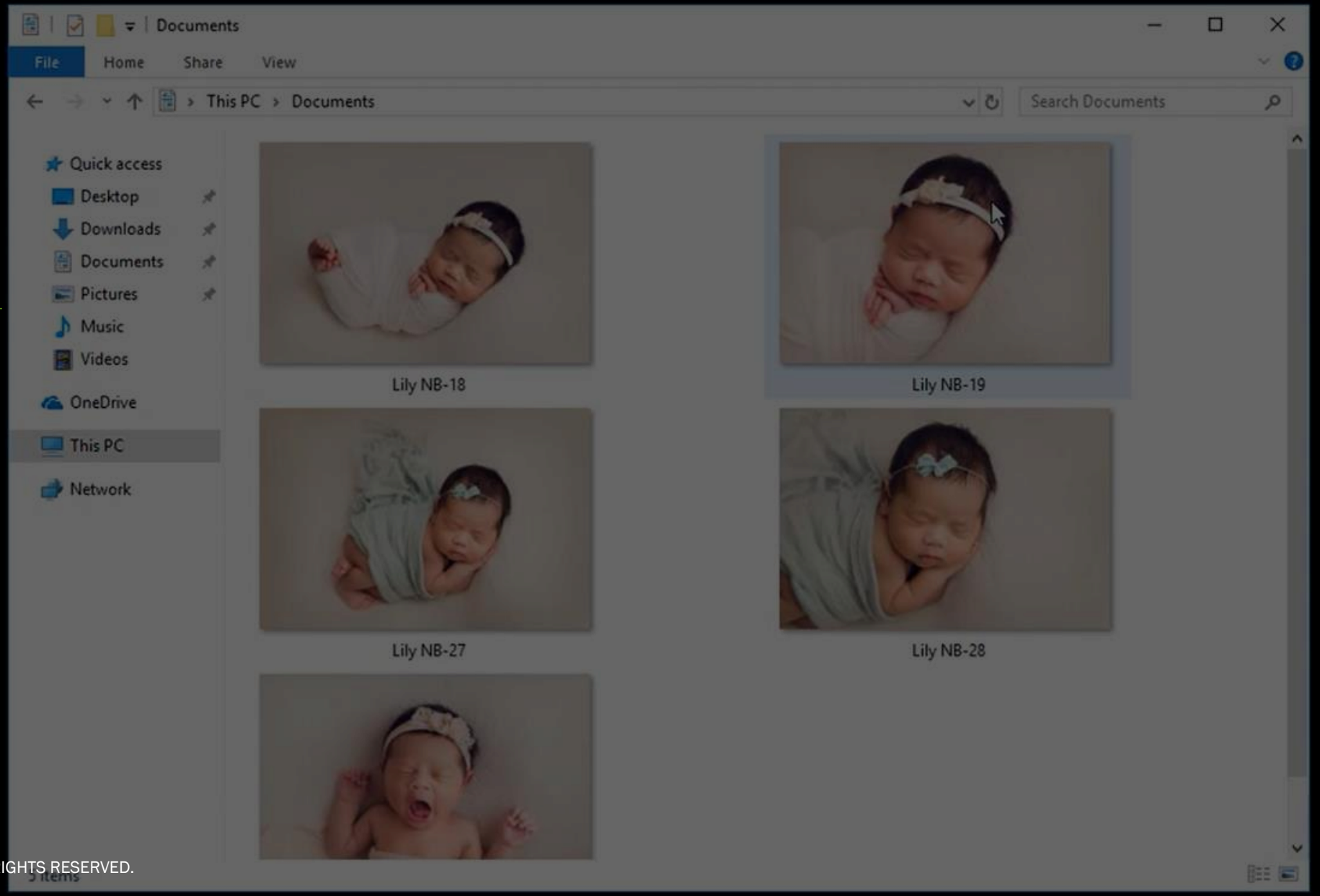
AGENDA



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DEMO

https://www.youtube.com/watch?v=OwJSZT2U4kM



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The Problem – Ransomware

A look at Ransomware Trends

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Timeline

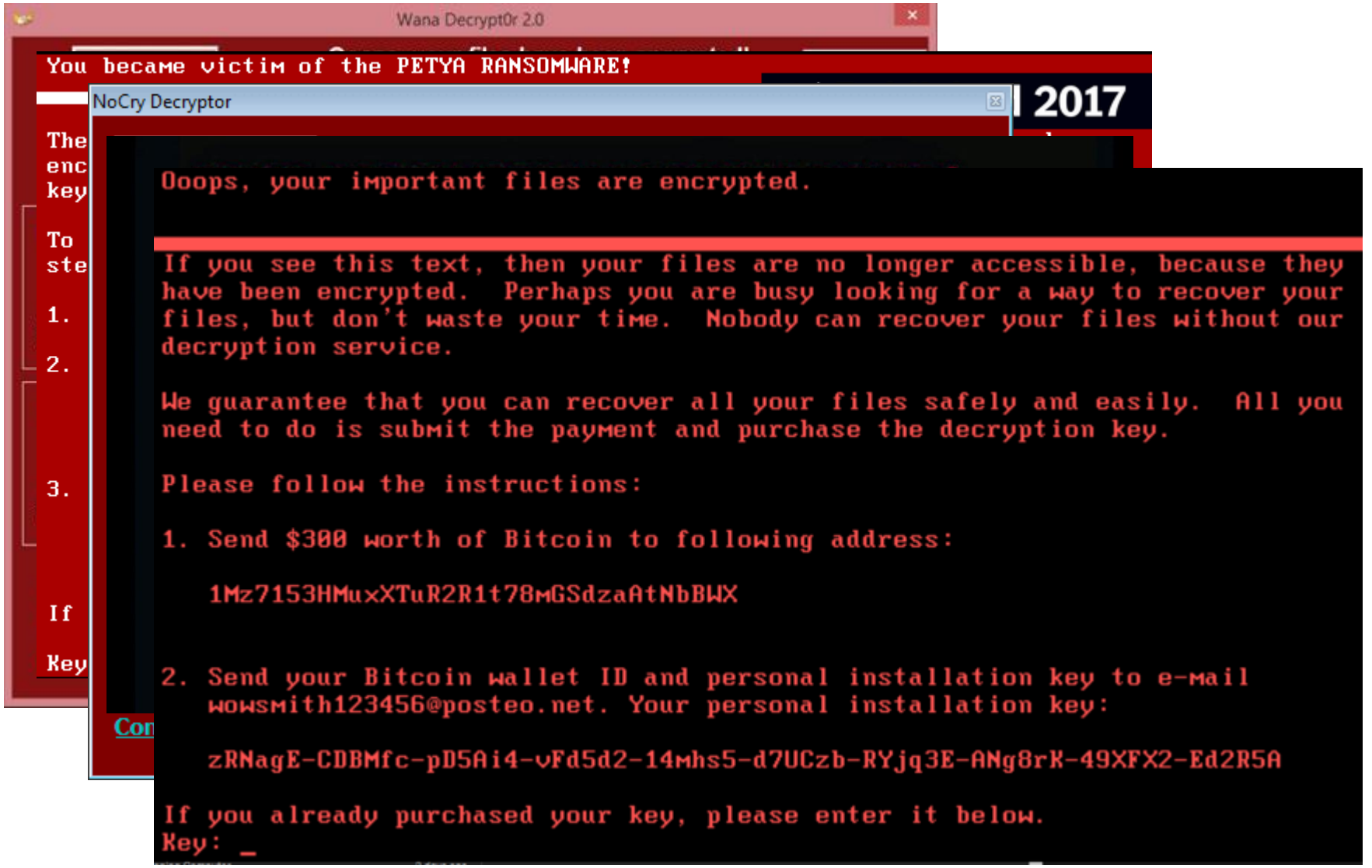


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Circa 2016

- Top 10 Worst Ransomware Attacks

1. NotPetya
2. WannaCry
3. Locky
4. CrySis
5. Nemucod
6. Jaff
7. Spora
8. Cerber
9. Cryptomix
10. Jigsaw



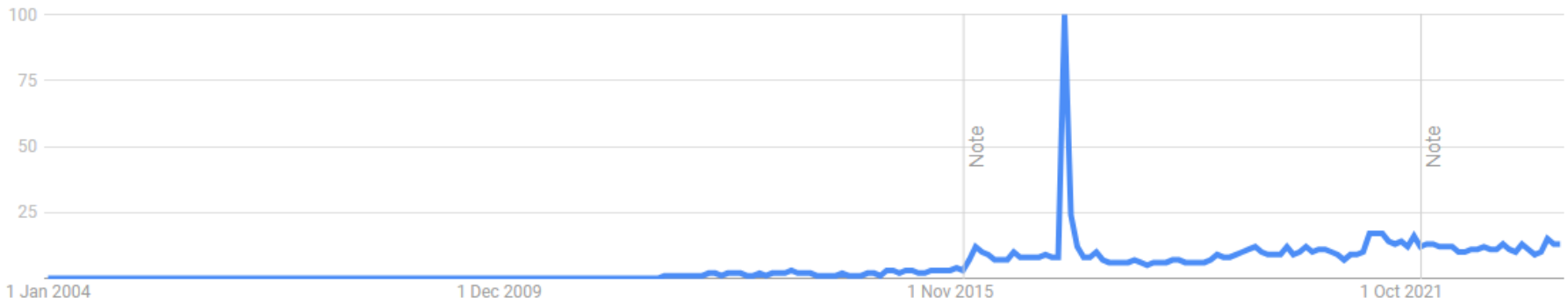
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ransomware
Search term

+ Compare

Worldwide | 01/01/2004 - 06/11/2023 | All categories | Web Search

Interest over time



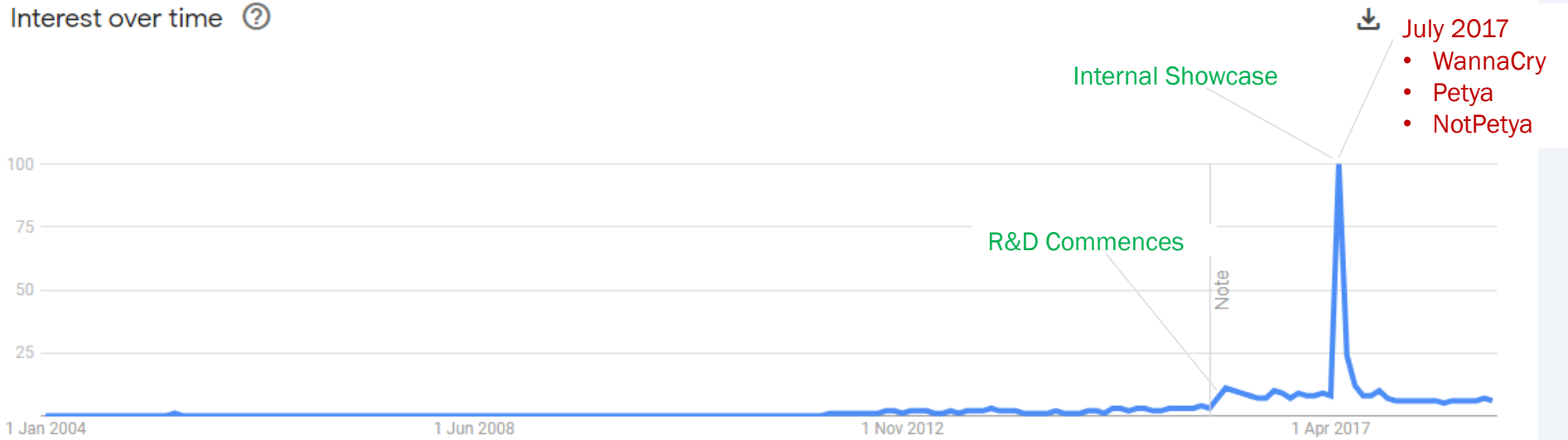
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ransomware
Search term

+ Compare

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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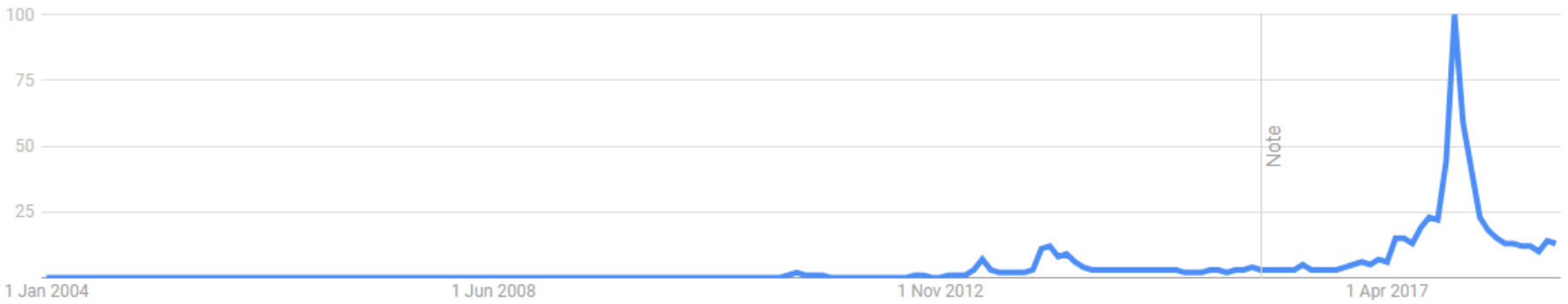
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Bitcoin
Search term

+ Compare

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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ransomware
Search term

Bitcoin
Search term

+ Add comparison

Worldwide | 01/01/2004 - 31/12/2018 | All categories | Web Search

Interest over time



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Ransomware continues to be a Threat

- US is the most attacked country.

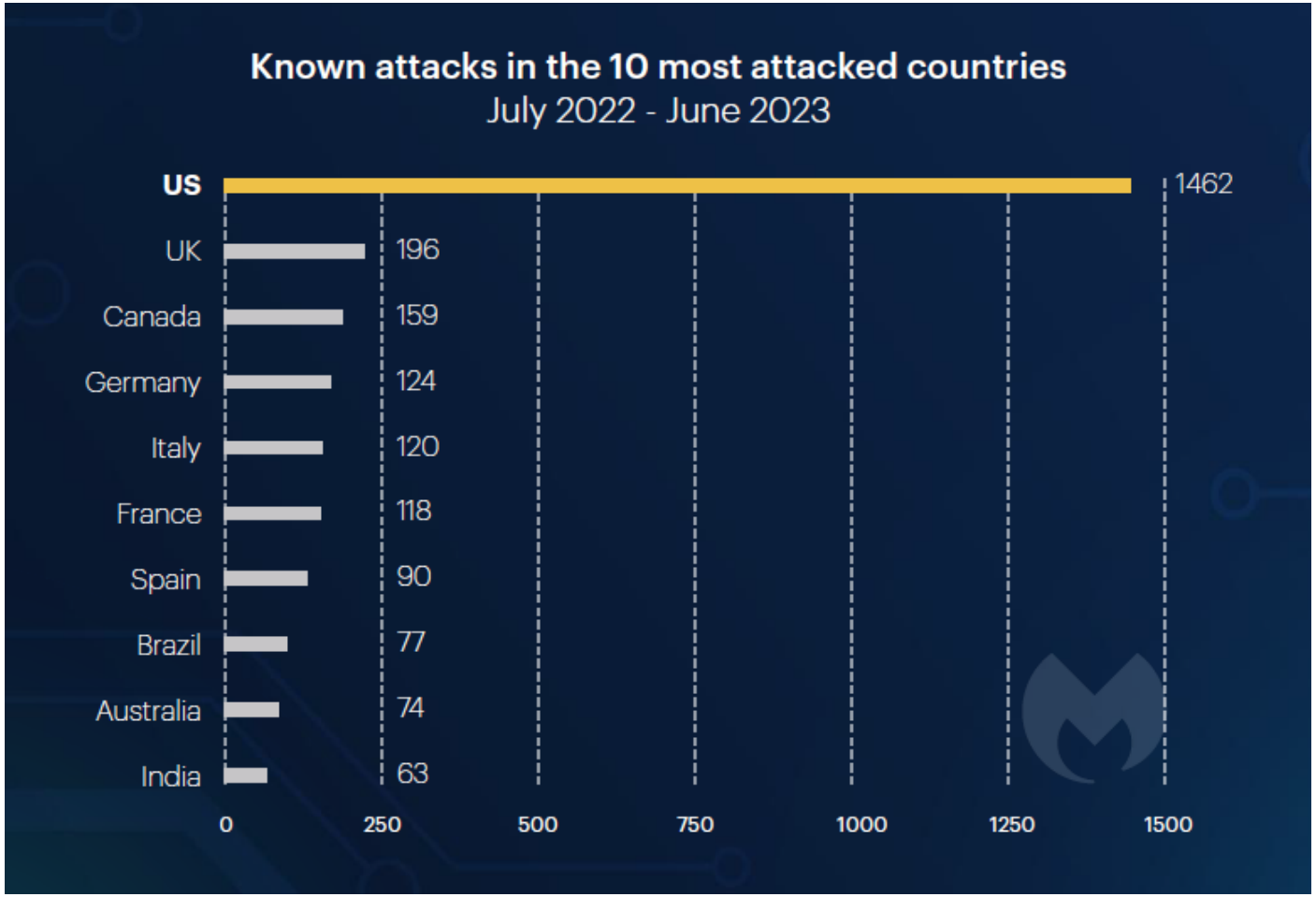


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Ransomware attacks in last 12 months.

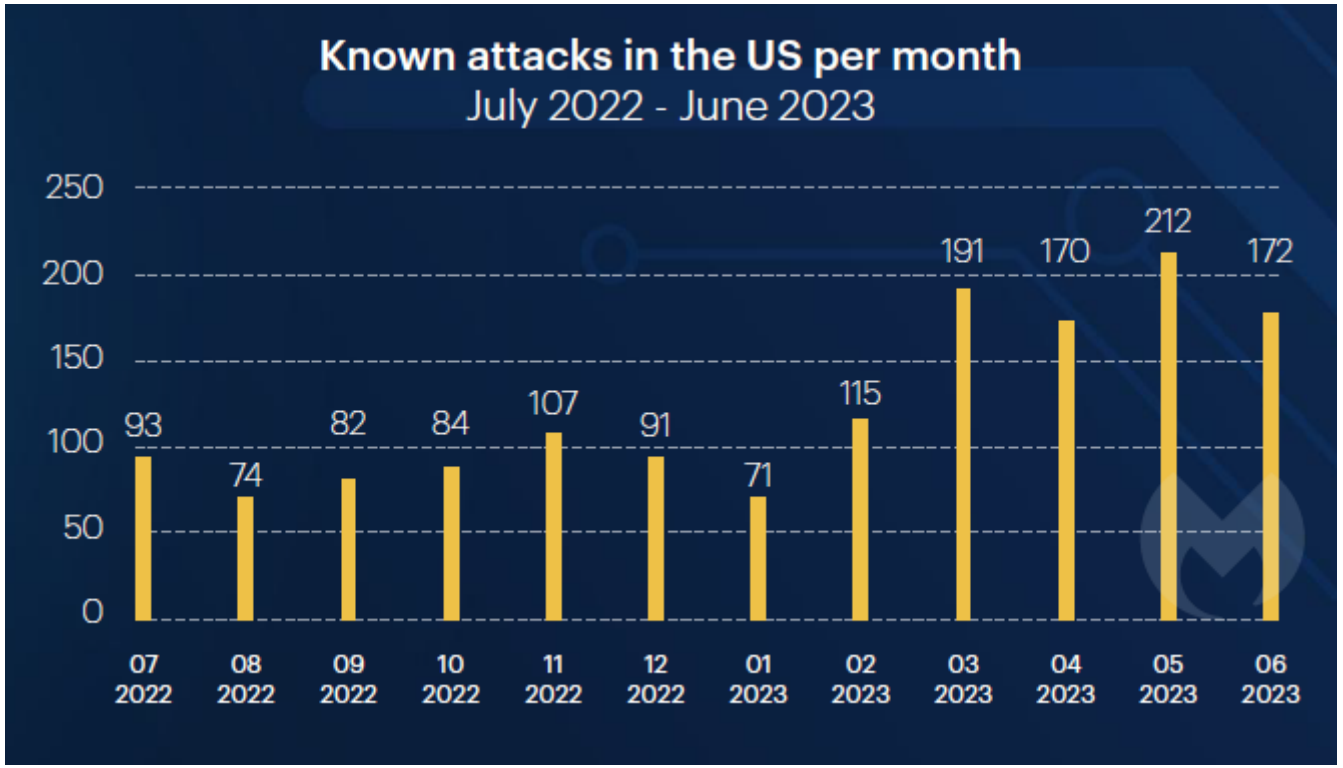


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

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Ransomware continues to be a Threat

- Active gangs

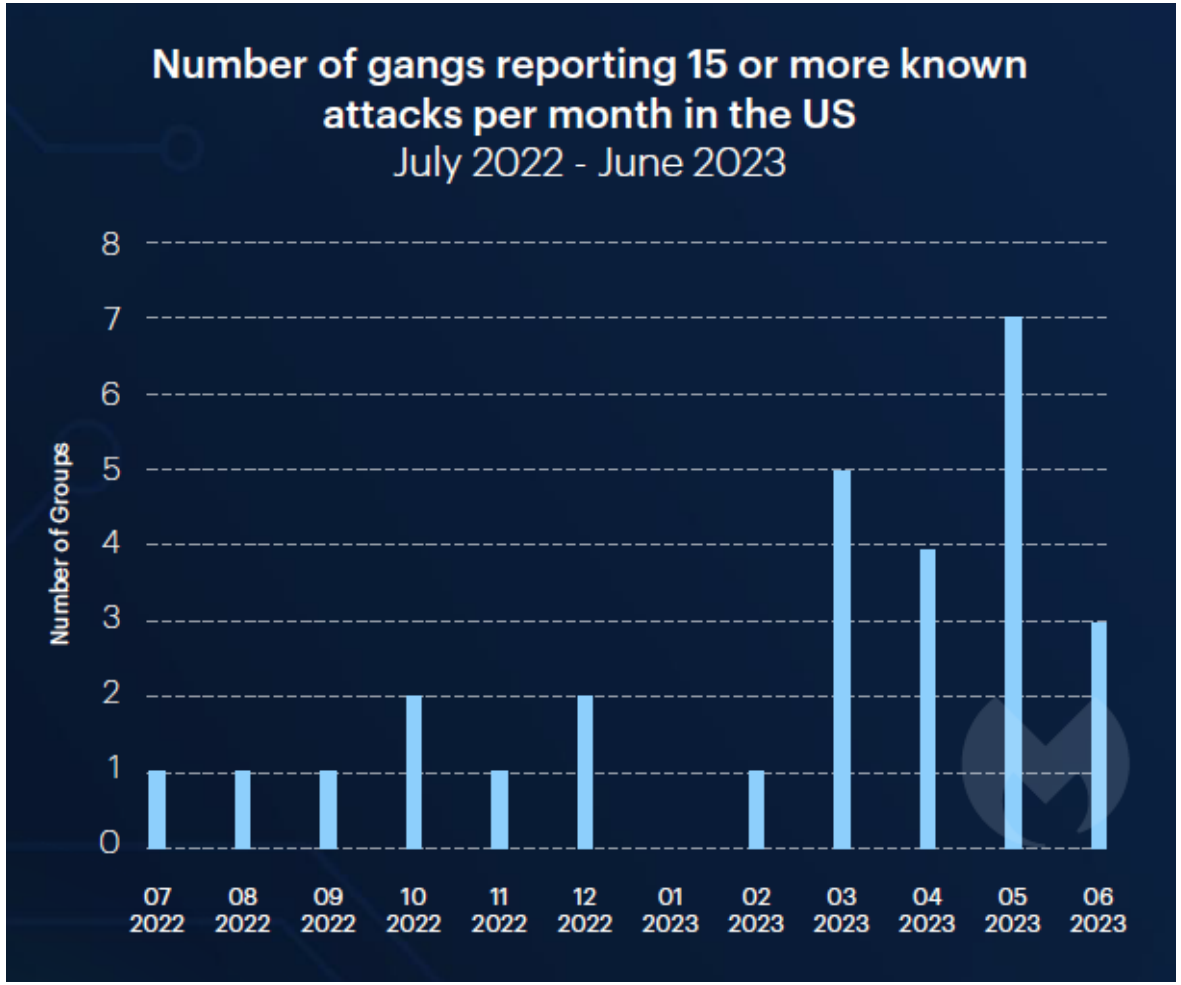


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Ransomware continues to be a Threat

- Ransomware Gangs

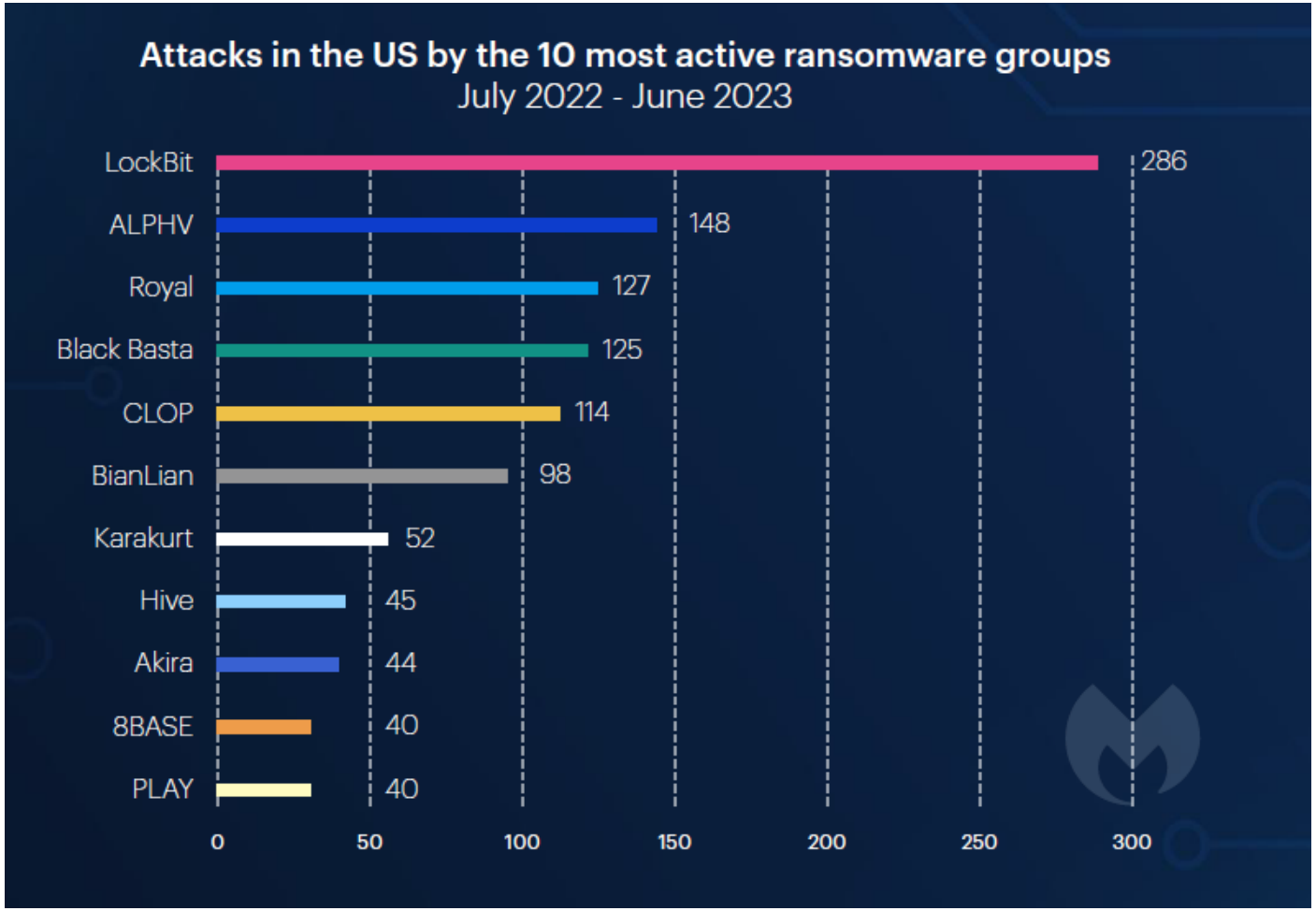


Image Credit: *Malwarebytes – State of Ransomware Report 2023*

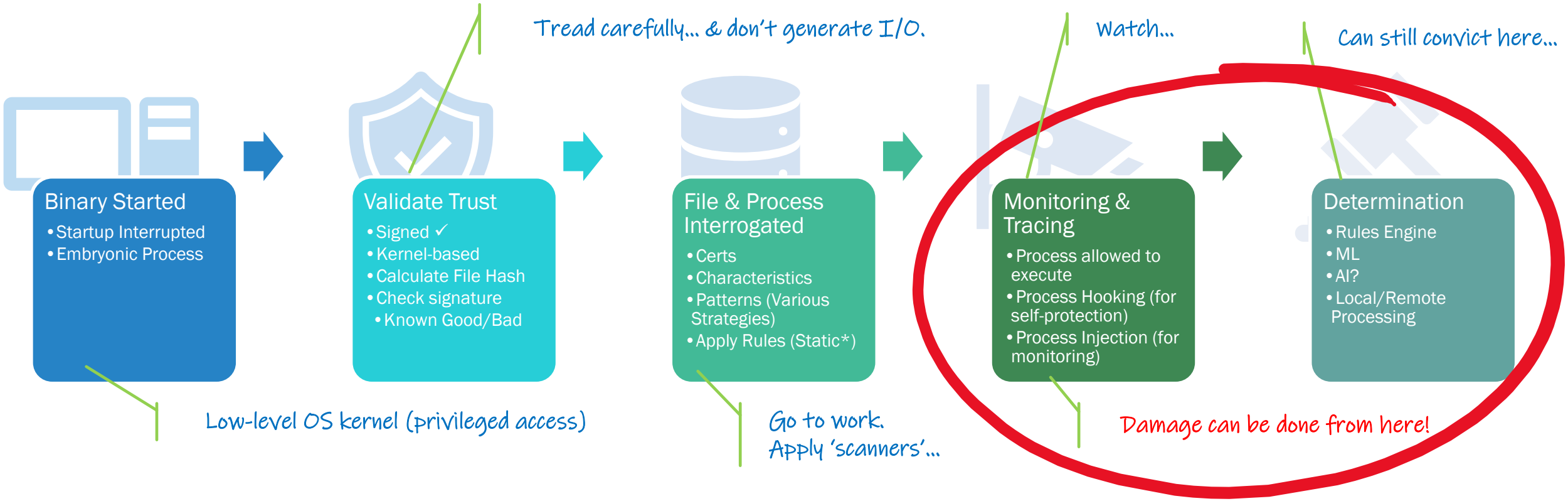
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Context – How AV Works

Typical Endpoint Antimalware

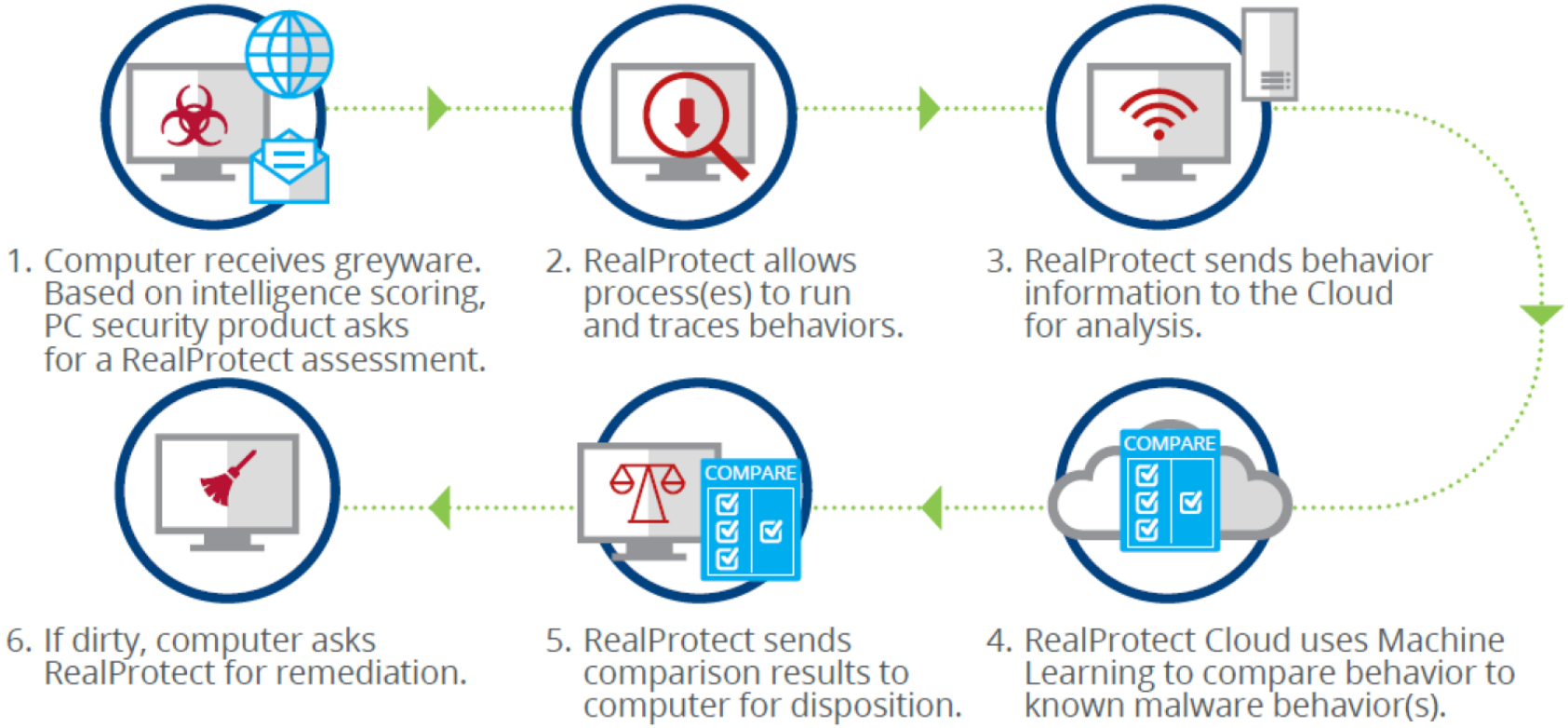
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How AV works



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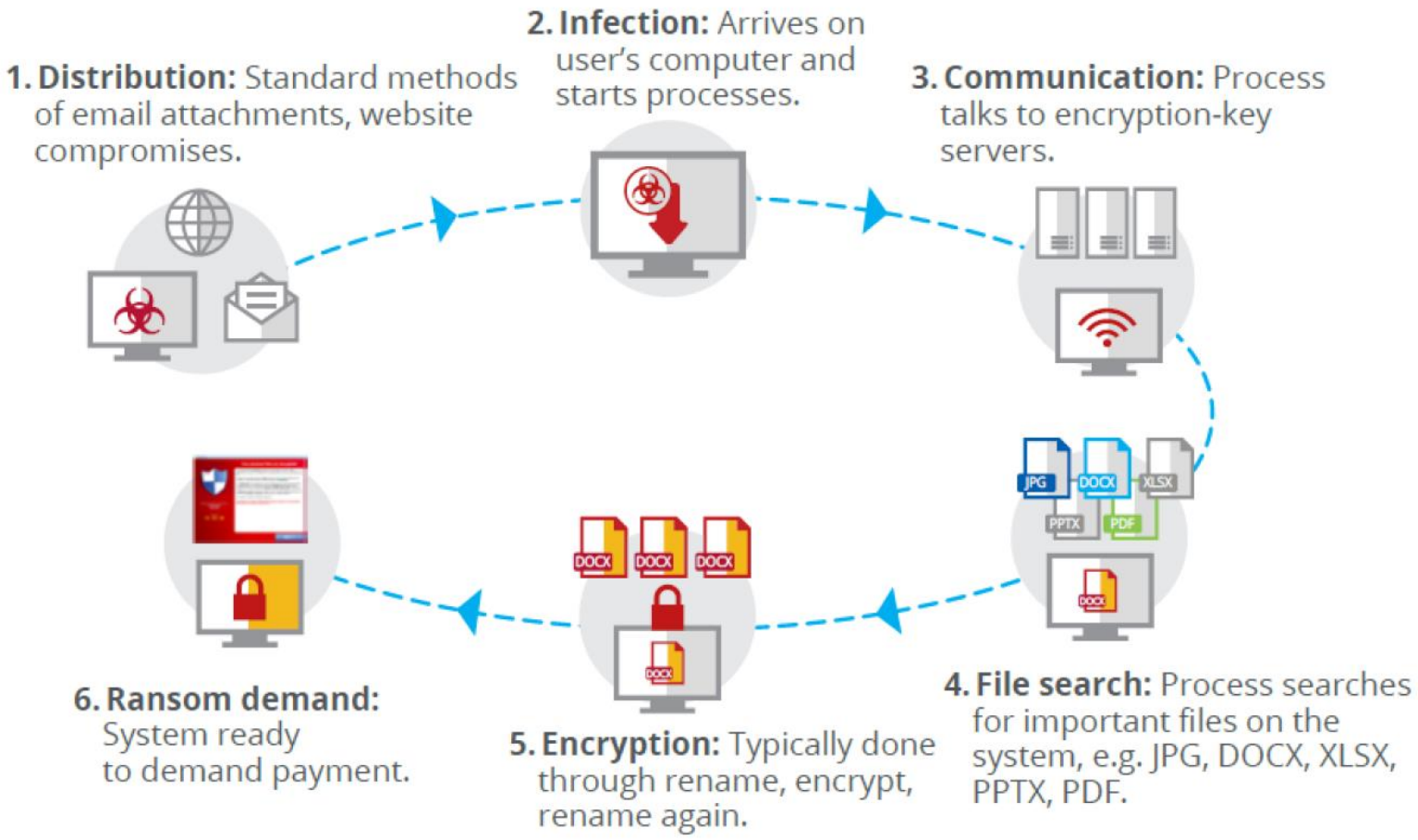
McAfee Behavioural Analysis



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Image Credit: McAfee Whitepaper – *You, Machine Learning & Zero-Day Malware*

How Ransomware Works

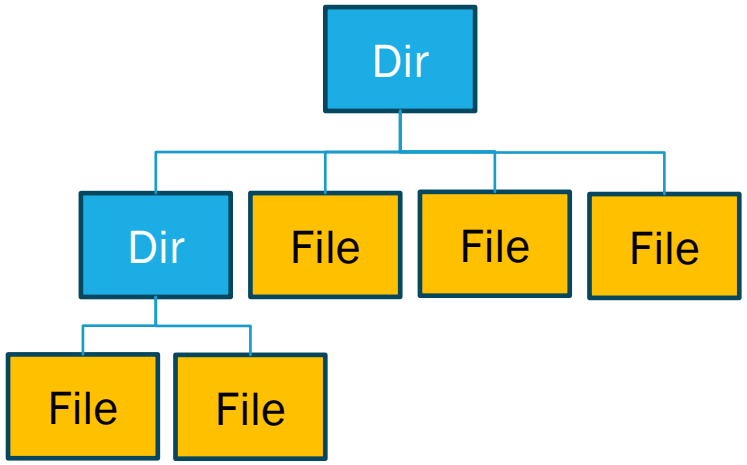


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Image Credit: McAfee Whitepaper – *You, Machine Learning & Zero-Day Malware*

How Ransomware Works

Iterate over the File System



*Focus on typical document locations
*Target office docs

Pseudo Code

Open File System

- Open Directory
- Read File
- Write Encrypted File
- Delete Original
- Loop

Loop

What's the Problem?

Zero-day!

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Innovation

Devising an Anti-ransomware Solution

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Strategies



Prevention

Block Malware

Zero-day Ransomware?

Polymorphic Malware?



Avoidance

Cloud Storage

(e.g. Microsoft OneDrive)

Minimize impact → Restore to last known-good backup

Incur re-image cost & effort

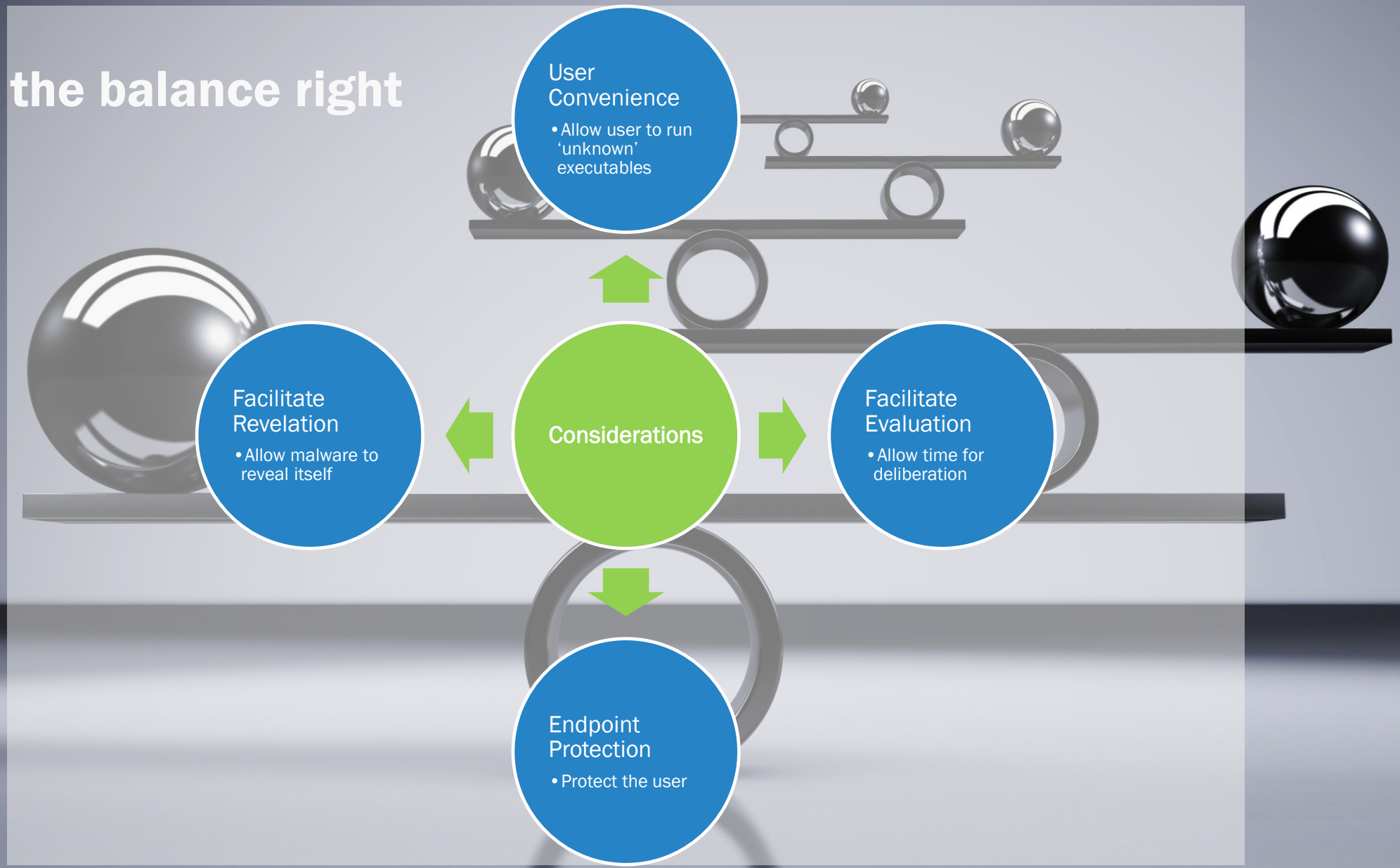


Remediate

Restore Endpoint to previous known-good state

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Getting the balance right



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REMEDIATION STRATEGIES



Confinement

Limit damage to a Virtual Image

Ditch VM on conviction

Roll forward if not convicted



Rollback Reversal

Perfect Undo

Rollback on conviction

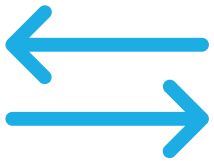
Do nothing if not convicted

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Remediation Strategies



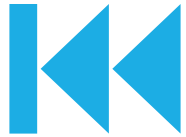
Optimistic v Pessimistic View



Roll forwards – or – Roll backwards



Most software != malware



Fewer instances where we need to take action → Roll backwards

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Rollback Remediation

What is it? How does it work?

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Rollback Remediation



Before Conviction

Monitor

Create Backups



Post Conviction

Kill Process Tree

Restore

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Rollback Remediation



Correction Technology

Does not detect

Does not protect



Passively Monitors

Performs Low-level Monitoring

Maintains ‘Process Genealogy’ Tree



Active Monitoring Triggered

Records detailed activity

Maintains copies of File, Directory and Registry changes.

Monitors child processes, proxies or any other surrogates.



Remediates

Terminates Process Tree

Uses Logs to restore all File, Directory & Registry changes & perform rollback.

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Remediation Details



File & Directory

Files & Directories restored to original state.

Encrypted files removed.

‘Dropped files’ removed.



Windows Registry

RegKeys & Values restored to original state.

Created RegKeys & Values removed.



File Attributes

Preserve Hidden/Archive file attributes.



Security Descriptors

Preserve file security metadata.

Restore File/Directory ownership.



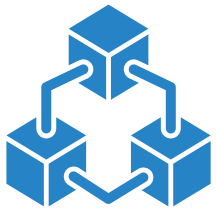
RunKeys & Startup Directories

Clean autostart registrations.

Clean known (and not-so-well known) directory & registry locations.

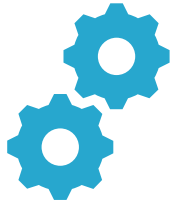
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Detecting Lateral Movement



Windows Service Control Manager

Registration of background Windows Services (Daemons).



Windows Task Scheduler

Remove tasks scheduled for future execution.



Windows BITS

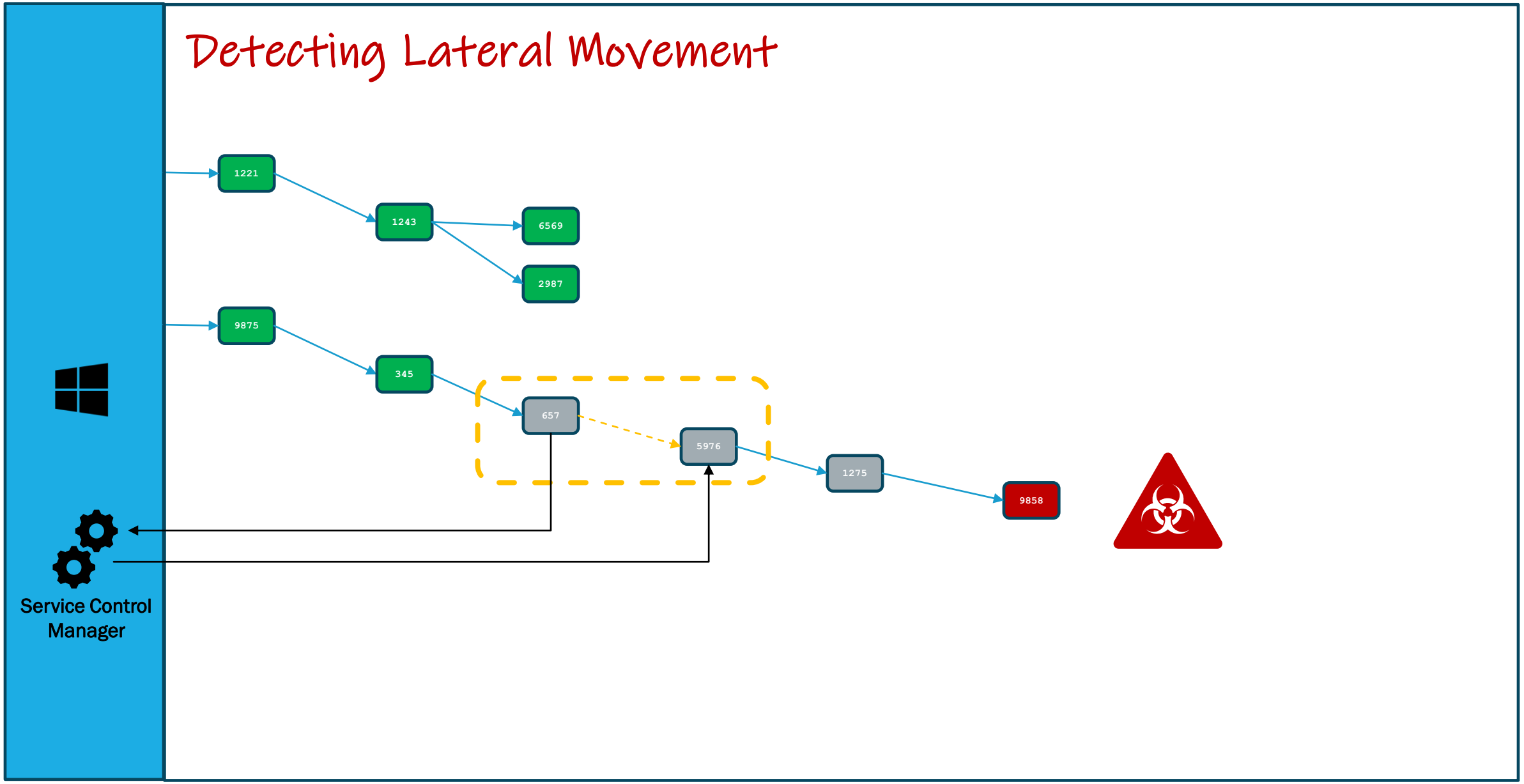
Background Intelligent Transfer Service

Facilitate intelligent file download

Download failure triggers registered handler.

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Detecting Lateral Movement



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Engineering

Building Solutions for GA

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ENGINEERING PRACTICES



Agile/Scrum

Daily Standup

2-week Sprint

6-week PSI (Potentially Shippable Increment)

End-of-Sprint Demo

Sprint Planning

Proxy Product Owner



Test Driven Development

Pair Engineers with SDET

Automated Functional Tests

Integrated into build

1000+ Test Cases



Continuous Build

Daily/Weekly/On-demand (PR)

Integrated Tests



Virus Labs

Test against specific Malware

Remediate Top-10 Ransomware

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Performance Engineering

Ensuring ‘cost’ is within bounds

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Performance Engineering



Endpoint is ‘highly’ sensitive to resource utilization

Active monitoring costs

e.g. File copies



EPP should be in the background

Operation imperceptible to the user



Public Tests rule the day

AV-TEST

AV-Comparisons

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Performance Engineering



Real-world Scenarios

Install Browser

Install LibreOffice

Uncompress operation



Quantitative Loads

Determine if performance is linear or potentially exponential

Variables = No. of Files, Size/Type of Files, Directory depth/breath

Scientific Approach – i.e. modify 1 variable at a time.



User Profile

Define typical profiles for light, medium & heavy users

e.g. Browsing & Streaming, MS Office user, Software Engineer



Windows ADK

Internal Standard

Common Use-cases

(e.g. browser load, application launch, boot-up, restart, etc.)

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Performance Engineering



Manage Performance Metrics

Capture & Store performance data

Graphically display results



Performance Trends

Track performance data over time.

Determine general improvements

Capture anomalies/outliers



CI/CT

Continuous Testing

Integrated into build

Hourly/daily/weekly builds

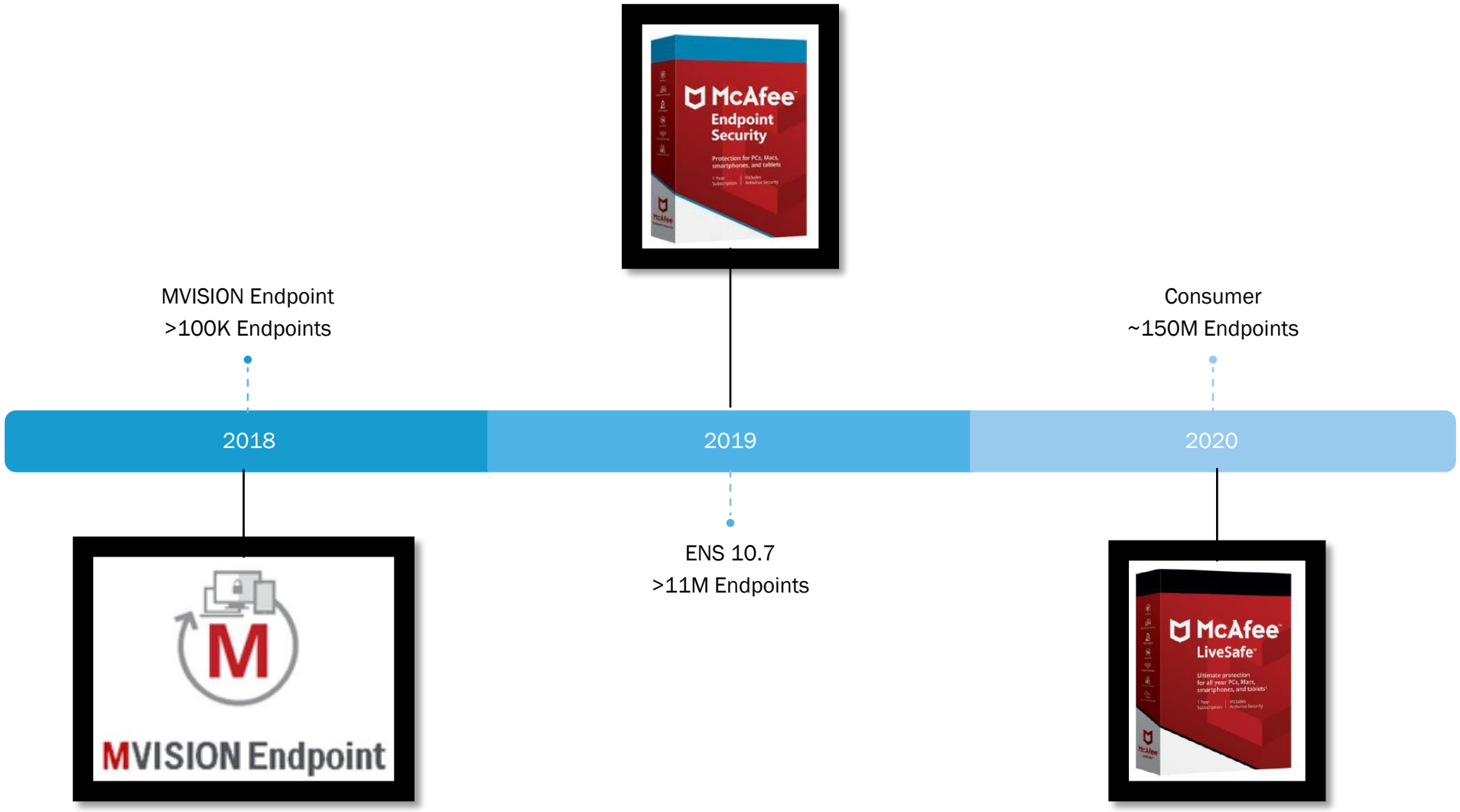
Show trend

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Conclusion

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Integration Timeline



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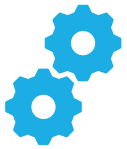
Conclusion



Ransomware Trends



How AV works



Innovation Journey



Engineering a Rollback Remediation solutions



Further Innovation…

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Potential Innovations



Deep Monitor ‘Greyware’



Track low-level operation of Malware AND Goodware



Big Data → Machine Learning → AI

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Thank you

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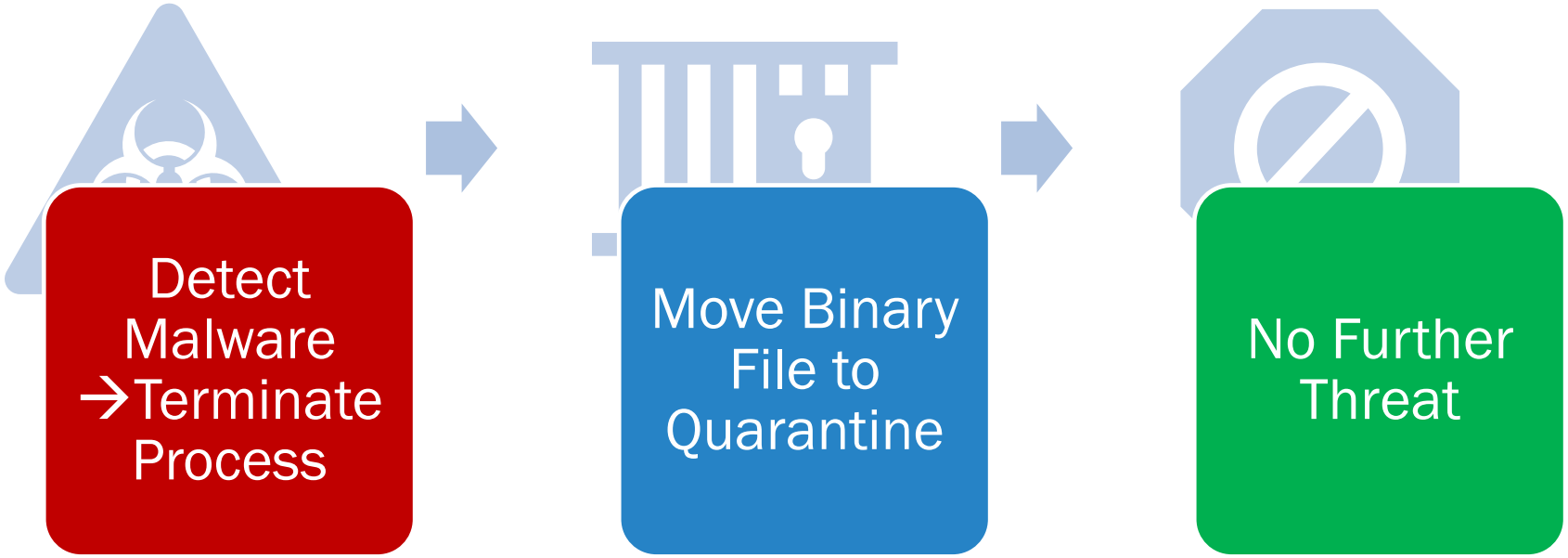
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Fileless Remediation

Breaking the Execution Chain

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Normal AV



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Systems

System Tree

Threat Event Log: Details — Previous | Next

Threat Handled:	True
Analyzer Detection Method:	MVISION Endpoint

Events received from managed systems

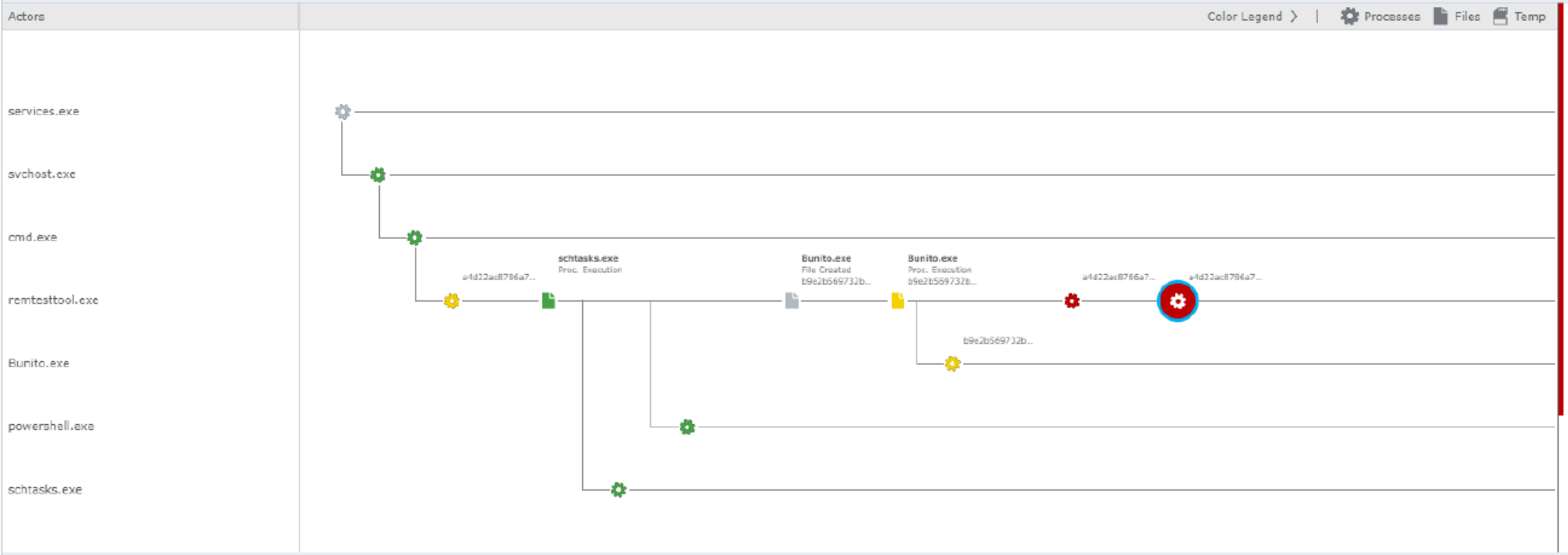
Event Description:	Infected file was moved to quarantine area.

Endpoint Security

Adaptive Threat Protection Events

Web Control Events

Story Graph (Trace Summary)



Event Details

Process terminated

Target Name	C:\Users\Ted\Desktop\deep_remediation_sample\remtesttool.exe
Reputation	Known Malicious
Reputation Score	1
PID	2964:132158147741768607

Action Taken
Threat has been quarantined

SHA-256
a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

MD5
4bf25e06e9e6c86e797da9a0cbc72fbb

Command Line Parameter
.\remtesttool.exe -sb -ib -wb -r -w1 18000 -db -rb -w2 180000

MVISION Endpoint

Target MD5 Hash:	4bf25e06e9e6c86e797da9a0cbc72fbb
Target SHA-256 Hash:	a4d22ac8786a75350f8ce8cc9f0f493fae227a6ff1e68945b23735f413785f49

Related Items

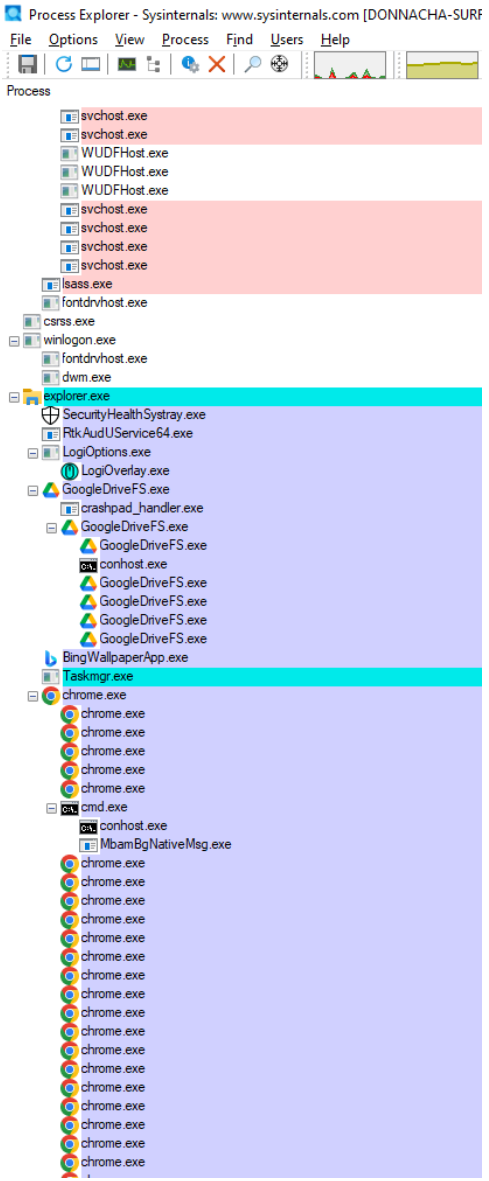
Go to related Systems

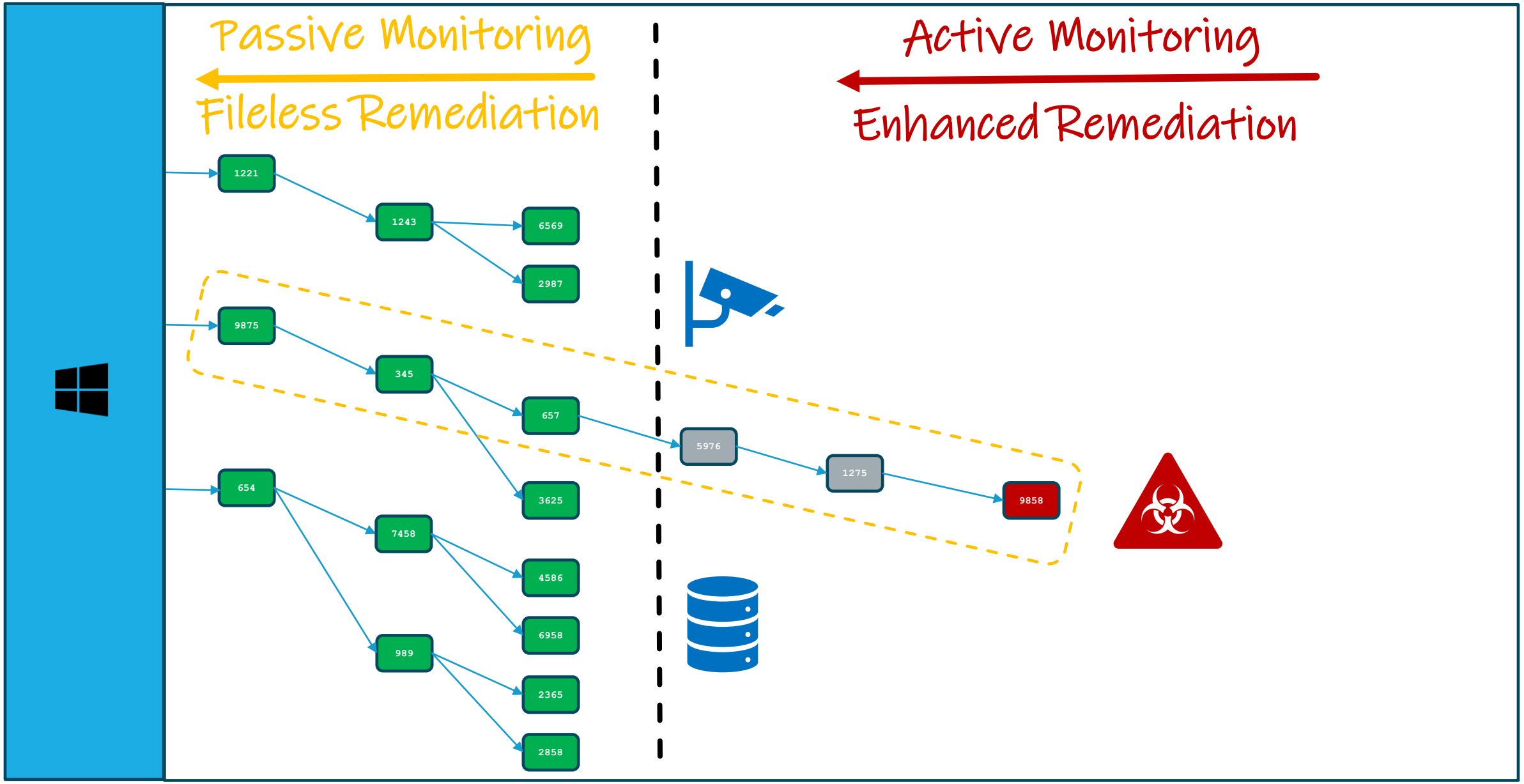
Actions | Show Source Systems | Show Target Systems

Back | Close

Process Tree Hierarchy

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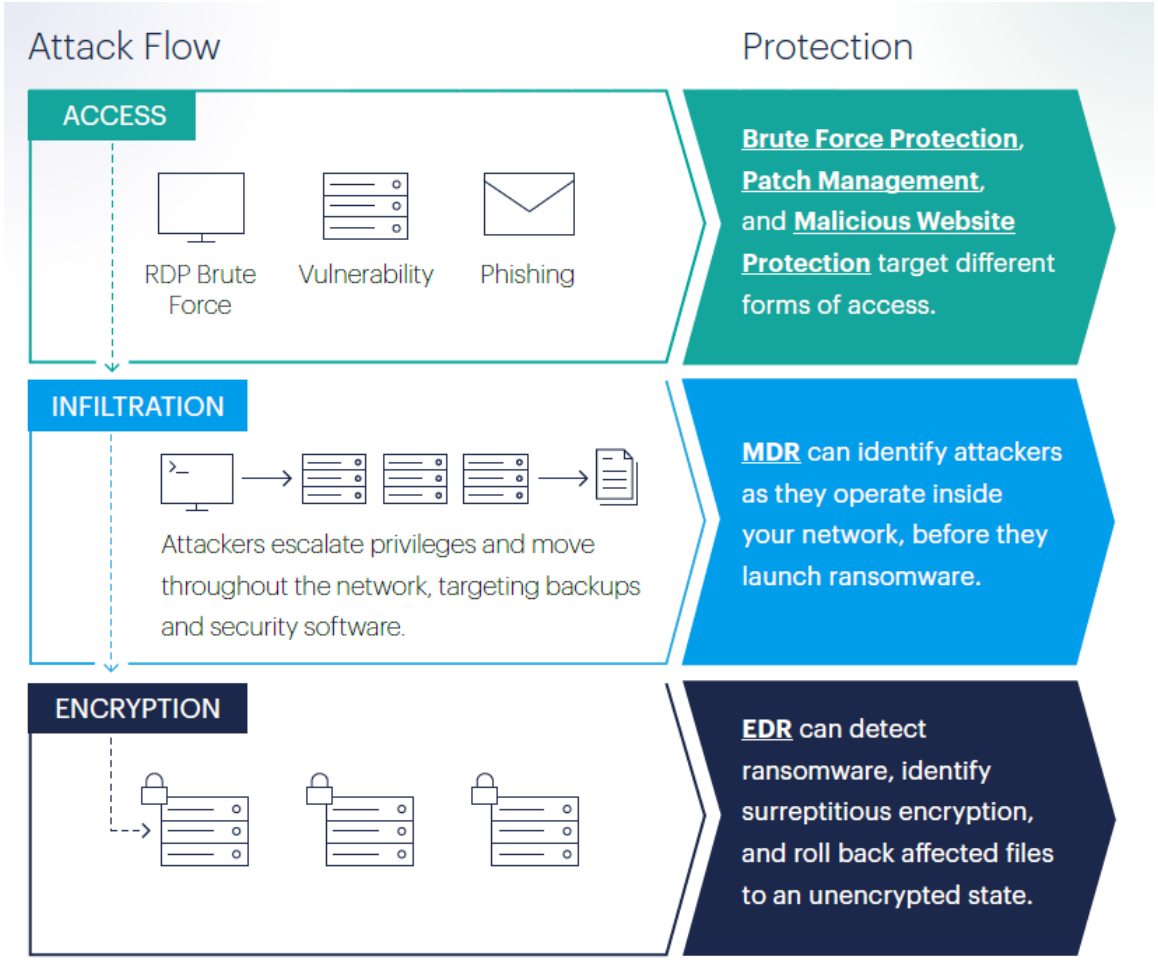


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A GEORGE PAL PRODUCTION
H.G. WELLS'
THE TIME MACHINE
in futuristic METROCOLOR
STARRING ROD TAYLOR · ALAN YOUNG · YVETTE MIMIEUX
SEBASTIAN CABOT · TOM HELMORE
Screen Play by DAVID DUNCAN · Based on the Novel by H. G. WELLS · Directed by GEORGE PAL

How Ransomware Works



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